

Environment variables and secrets

Module 6 - Lesson 1

Overview

Configuration differs between environments, and secrets must never reach the client or the repository. Environment variables separate config from code.

Key Concepts

- .env files hold local config and are excluded from git (gitignore).
- Prefix VITE_ to expose a variable to a Vite client build; everything else stays server-only.
- Secrets like API keys and database URLs belong in the deployment platform's settings, not in code.
- Validate that required variables exist at startup and fail fast with a clear message.
- Never log secrets; redact them in request logs.
- Use a single source of truth for config to avoid drift between environments.

Example

```
// .env (never committed)
DATABASE_URL=postgres://...
JWT_SECRET=change-me-now

// server only
const secret = process.env.JWT_SECRET;
// never accessible in the browser bundle
```

Summary

Environment variables keep secrets out of code and out of the client. Validate required variables at boot, keep client-facing vars explicitly prefixed, and never log secrets.

Practice Checklist

- Move hard-coded credentials in your app into environment variables.
- Verify no secrets appear in the built client bundle.
- Add startup validation for all required variables.